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Eser Aygün, Anastasiya Belyaeva, Gheorghe Comanici, Marc Coram, Hao Cui, Jake Garrison, Renee Johnston, Anton Kast, Cory Y. McLean, Peter Norgaard, Zahra Shamsi, David Smalling, James Thompson, Subhashini Venugopalan, Brian P. Williams, Chujun He, Sarah Martinson, Martyna Plomecka, Lai Wei, Yuchen Zhou, Qian-Ze Zhu, Matthew Abraham, Erica Brand, Anna Bulanova, Jeffrey A. Cardille, Chris Co, Scott Ellsworth, Grace Joseph, Malcolm Kane, Ryan Krueger, Johan Kartiwa, Dan Liebling, Jan-Matthis Lueckmann, Paul Raccuglia, Xuefei Julie Wang, Katherine Chou, James Manyika, Yossi Matias, John C. Platt, Lizzie Dorfman, Shibl Mourad & Michael P. Brenner

An AI system to help scientists write expert-level empirical software

Eser Aygün[1,*], Anastasiya Belyaeva[2,*], Gheorghe Comanici[1,*], Marc Coram[2,*], Hao Cui[2,*], Jake Garrison[3,*], Renee Johnston[2,*], Anton Kast[2,*], Cory Y. McLean[2,*], Peter Norgaard[2,*], Zahra Shamsi[2,*], David Smalling[1,*], James Thompson[2,*], Subhashini Venugopalan[2,*], Brian P. Williams[2,*], Chujun He[2,4,**], Sarah Martinson[2,5,**], Martyna Plomecka[2,6,**], Lai Wei[2], Yuchen Zhou[2], Qian-Ze Zhu[2,5,**], Matthew Abraham[2], Erica Brand[2], Anna Bulanova[1], Jeffrey A. Cardille[2,7], Chris Co[2], Scott Ellsworth[2], Grace Joseph[2], Malcolm Kane[2], Ryan Krueger[2,5,**], Johan Kartiwa[2], Dan Liebling[2], Jan-Matthis Lueckmann[2], Paul Raccuglia[2], Xuefei (Julie) Wang[2,8,**], Katherine Chou[2], James Manyika[2], Yossi Matias[2], John C. Platt[2], Lizzie Dorfman[2], Shibl Mourad[1,‡], and Michael P. Brenner[2,5,‡]

[1] *Google DeepMind, Montréal, Quebec H2Z 1W5, Canada*
[2] *Google Research, Cambridge, MA 02142, USA*
[3] *Google Platforms and Devices, Mountain View, CA 94043, USA*
[4] *Massachusetts Institute of Technology, Cambridge, MA 02139, USA*
[5] *School of Engineering and Applied Sciences, Harvard University, , Cambridge, MA 02138, USA*
[6] *Google DeepMind, New York, New York 10011, USA*
[7] *Faculty of Agricultural and Environmental Sciences, McGill University, Montréal, Quebec H3A 0G4, Canada*
[8] *California Institute of Technology, Pasadena, CA 91125*

[*] Equal contribution in alphabetical order.
[**] Carried out as part of a student researchership at Google Research.
[‡] To whom correspondence should be addressed: shibl@google.com, mbrenner@google.com

The cycle of scientific discovery is frequently bottlenecked by the slow, manual creation of software to support computational experiments[1]. To address this, we present Empirical Research Assistance (ERA), an AI system that creates expert-level scientific software whose goal is to maximize a quality metric. The system uses a Large Language Model (LLM) and Tree Search (TS)[2] to systematically improve the quality metric and intelligently navigate the large space of possible solutions. ERA achieves expert-level results when it explores and integrates complex research ideas from external sources. The effectiveness of tree search is

demonstrated across a diverse range of tasks. In bioinformatics, ERA discovered 40 novel methods for single-cell data analysis that outperformed the top human-developed methods on a public leaderboard. In epidemiology, ERA generated 14 models that outperformed the CDC ensemble and all other individual models for forecasting COVID-19 hospitalizations. ERA also produced expert-level software for geospatial analysis, neural activity prediction in zebrafish, and numerical solution of integrals, and a novel rule-based construction for time series forecasting. By devising and implementing novel solutions to diverse tasks, ERA represents a significant step towards accelerating scientific progress.

Keywords: Tree Search, Generative AI, Scorable Scientific Tasks, Empirical Software

Introduction

Empirical software, designed to maximize a measurable quality score, is ubiquitous and central to many scientific endeavors. Empirical software has recently enabled a number of Nobel Prizes in Chemistry: in 1998 for Density Functional Theory[3,4], in 2013 for molecular dynamics simulation[5] and in 2024 for protein structure prediction[6,7]. Empirical software underlies our ability to create models of complex systems, ranging from parameterizations of a vertical column of the earth's atmosphere for weather modeling[8], to the parameterization of stress response in a turbulent fluid flow[9], to the prediction of social systems[10,11].

However, *empirical software for science is slow and difficult to create*. Domain-specific empirical software requires tedious work[1], often over many years. When empirical software is used to test complex hypotheses, it becomes ever more difficult to write purely from first principles. There usually is no systematic search for alternative approaches. Design choices are often governed by intuition or expediency, rather than exhaustive experimentation. Creating the software is so time-consuming that it severely limits the possibilities that can be productively explored[12].

Here we present Empirical Research Assistance (ERA), an AI-based system that systematically and automatically creates empirical software to solve scorable tasks. ERA is based on an LLM that rewrites software to attempt to improve its quality score. ERA creates multiple software candidate solutions, and uses Tree Search[2] to decide which candidates merit further exploration (Fig. 1a). While there are many ways of designing a code mutation system[13–16], we developed ERA by competing in Kaggle competitions (Fig. 1b), described below. We augment code mutation with research ideas obtained from a range of sources including highly-cited papers, specialized textbooks, and search engine results (Fig. 1c). In practice, these ideas can be injected either directly by the user or automatically using a search engine to access research in the literature. The LLM uses this injected guidance in writing code.

We find that ERA produces software that outperforms the state-of-the-art in scorable tasks spanning broad scientific disciplines. This expert-level performance arises because of the ability to exhaustively and tirelessly carry out solution searches at unprecedented scale, identifying needle-in-the-haystack high quality solutions.

Results

Overview of Scorable Tasks

We develop ERA on Kaggle playground competitions, and test it by selecting scorable tasks based on scientific or engineering problems of high relevance in diverse fields. These scorable tasks are listed below, with per-node computational costs outlined in Supplementary Table S1.

scRNA-seq batch integration:[17] This task requires distinguishing subtle biological signals from noise in high-dimensional sparse datasets. By removing confounding factors, we can enable large-scale multi-lab transcriptomic data integration.

CDC COVID Forecasting:[18] This task requires predicting non-linear disease dynamics from lagged and noisy real-time data. By predicting COVID cases several weeks in advance, we can inform public health policy and resource allocations.

Time series forecasting:[19] This task requires predicting time series outcomes across a range of datasets and applications.

Geospatial segmentation:[20] This task requires performing dense pixel-wise multi-label semantic segmentation on complex satellite imagery. Better segmentation can lead to large improvements in environmental monitoring and disaster response.

ZAPBench:[21] This task requires predicting the activity of >70,000 neurons across an entire vertebrate brain. Performing well on this benchmark may lead to a systems-level understanding of brain function and behavior.

Numerically solving difficult integrals: This task requires solving integrals that defy standard numerical algorithms. It is useful for modeling physical and engineering systems.

Kaggle Playground Benchmark

We designed ERA to score highly on a curated set of Kaggle competitions. Kaggle calibrates human performance with percentile rank on a leaderboard, and we score code by submitting directly to Kaggle. Our benchmark consists of 16 playground competitions from the 2023 season, encompassing regression and classification tasks (Supplementary Table S2). Playground competitions are an ideal benchmark because they offer fast iteration, simplicity, and calibration against thousands of humans. Achieving a high score requires creating complex code without requiring solving a sophisticated scientific task.

Our basic strategy uses a simple prompting template (Supplementary Table S3) that concatenates the competition description with the previous trial. Fig. 1b evaluates the performance of ERA with the average public percentile rank across all 16 playground competitions: ERA substantially beats a single LLM call and best-of-1000 LLM calls, and also outperforms AIDE[13], owing to its ability to maintain a diverse tree of candidates, allowing it to backtrack when a specific line of code mutation plateaus. During the search, the system discovers strategies leading to abrupt jumps in the score, with the accumulation of these jumps leading to the highest quality solutions.

Problem-specific advice added to the prompt substantially improves performance. We illustrate this with two examples. In *TS with expert advice* we give ERA standard advice to win Kaggle competitions (Supplementary Table S4). In *TS with Boosted Decision Tree (BDT)* we tell ERA to implement a boosted decision tree library from scratch, without

using standard packages (Supplementary Table S5). We manually verified in both cases that resulting codes followed the advice.

We now evaluate ERA on six benchmarks in different scientific fields, exploring distinct ways to incorporate research ideas to improve system performance (Fig. 1c, Methods).

Genomics: Batch Integration of Single-Cell RNA Sequencing Data

Single-cell RNA sequencing (scRNA-seq) has revolutionized our ability to dissect cellular heterogeneity, discover novel cell types, infer gene regulatory networks and developmental trajectories, and improve therapeutic target prioritization[22], enabling hundreds of millions of cells to be individually sequenced within thousands of datasets[23–25]. A major challenge required to jointly analyze many disparate scRNA-seq datasets is to computationally remove complex batch effects present across samples while preserving biological signals[26]. Nearly 300 tools exist to perform batch integration of scRNA-seq data[27], and multiple benchmarks have been developed for assessing metrics of batch effect removal and conservation of biological variability[28–30].

To assess ERA performance on this task, we used the OpenProblems v2.0.0 batch integration benchmark[30]. As of July 2025, this active benchmark evaluates 15 state-of-the-art methods and eight control methods on 13 different metrics that quantify both the ability to remove batch effects in the data and retain variability attributable to true biological differences in six datasets spanning human and mouse[25] (Fig. 2a). To avoid overfitting to the benchmark, we used a separate dataset for ERA optimization (Methods, Supplementary Fig. S1). For each ERA run, we selected the best solution based on the performance on this training set, and report the performance on the holdout OpenProblems datasets (n=1,747,937 total cells). We prompt the LLM with a description of the single cell batch integration problem, code for reading in the dataset, code for evaluation metrics, and optional text with a particular research idea.

First, we ran ERA without guidance, and observed that its solution is conceptually similar to ComBat[31], yet improved over the current OpenProblems leaderboard (`No advice (TS)` in Fig. 2b). We then evaluated whether ERA could improve upon existing algorithms. We selected nine methods from the OpenProblems benchmark, including the six highest-performing methods (Methods). For each method, we obtained the paper PDF and used Gemini 2.5 Pro to add a brief summary to the prompt (Methods). In pairwise comparisons, ERA outperformed the corresponding published result for eight of the nine methods in overall score (Fig. 2b). The top-performing method was an ERA implementation of Batch Balanced K-Nearest Neighbors (`BBKNN (TS)`)[32], yielding a 14% overall improvement over the best published method (`ComBat`[31]) and equaled or outperformed the corresponding published `BBKNN` in every dataset and across 11/13 metrics (Fig. 2b). This performance highlights its capacity to effectively remove batch effects without compromising biological signals (Supplementary Fig. S2). We observed that ERA is also able to produce performant implementations for an algorithm without publicly-available code (TabVI[33], Extended Data Fig. 1). Importantly, expert manual inspection of the code solutions proposed by ERA confirmed that nearly all implementations adhered to the requested algorithms (Extended Data Table 1), with performance largely consistent across replicate runs of methods (Extended Data Fig. 1). Additionally, ERA demonstrated improvements even when compared to base

methods with optimized hyperparameters, indicating that its contribution extends beyond hyperparameter tuning (Methods, Supplementary Fig. S3). Extended Data Fig. 2 shows the tree structure and evolution of the maximum score as a function of the number of nodes in the tree for the best performing model `BBKNN (TS)`.

For `BBKNN (TS)`, part of the performance boost came from combining two existing methods, ComBat[31] and BBKNN, rather than simply implementing BBKNN (Fig. 2c). In particular, while the original BBKNN method computes neighbors using PCA embeddings, `BBKNN (TS)` computes neighbors using ComBat-corrected PCA embeddings, removing global linear batch-associated variance. Both implementations then compute k-nearest neighbors across batches and construct a graph (with differences in exact implementation), thus removing local batch effects (Supplementary Table S6). Manual modification of `BBKNN (TS)` and the published BBKNN implementation confirmed that using Combat-corrected PCA embeddings is critical for improving both implementations (Extended Data Fig. 3), confirming the value in idea recombination.

This motivated an exploration of systematic ways to generate more complex research ideas. First, similar to how scientists often combine ideas to create a novel approach, we programmatically generated 55 "recombinations" of all pairs of the 11 methods described above (No advice, nine replications, and TabVI; hereafter: "base methods") based on summaries of the code for each method (Methods, Supplementary Table S7). We ran ERA, prompted with each of these recombinations. For each base method and recombination group, we compared the average scores for the top nodes over the intersection of metrics that were successfully computed for all three methods. Strikingly, recombination implementations of tree search frequently outperformed their base counterparts, with 24 of the 55 recombination solutions (44%) outperforming both of their base methods and 22 of the remaining 31 recombination solutions outperforming one of the two base methods (Extended Data Fig. 4). Second, we used Gemini Deep Research[34] and AI co-scientist[35] to generate and implement 21 additional ideas (Methods). After ERA was applied to each, 6/11 base methods, 29/55 recombination, 4/9 Deep Research, and 1/12 AI co-scientist methods (40 of 87) outperform all methods currently published on the OpenProblems leaderboard (Fig. 2d).

To further understand the conceptual space explored by ERA, we obtained embeddings for each generated code using Gemini text embedding model and computed cosine similarities (Supplementary Fig. S4). Visualization of the embeddings revealed distinct clusters, generally representing deep-learning based methods and non-deep-learning methods, suggesting that ERA is able to generate a variety of solutions (Supplementary Fig. S5).

Public Health: Prediction of U.S. COVID-19 Hospitalizations

The primary U.S. benchmark for COVID-19 forecasting is the COVID-19 Forecast Hub (CovidHub)[18], a large, collaborative effort coordinated by the Centers for Disease Control and Prevention (CDC). CovidHub receives weekly forecasts from dozens of expert-led teams across academia, industry, and government, each using different methodologies. These weekly forecasts must cover new COVID-19 related hospitalizations across 52 U.S. states and territories for the current week and three subsequent weeks over 23 specified quantiles. Submissions are evaluated using the Weighted Interval Score (WIS), which rewards both accuracy and well-calibrated uncertainty.

Top-performing individual models include classic autoregressive time-series approaches (e.g., UMASS-ar6_pooled), gradient boosting machine learning models (e.g., UMASS-gbqr), and epidemiological models based on renewal equations and Bayesian estimation of the reproductive number (e.g., CEPH-Rtrend_covid). CovidHub leverages this methodological diversity by integrating submissions into the CovidHub Ensemble, a robust aggregate forecast that has historically provided the gold standard for epidemiological prediction in the U.S.

We designed a rigorous retrospective study to assess ERA performance in this competitive environment, using data available on May 1 2025. For every forecasting period, we ran ERA to optimize and select a model using data from the preceding six weeks, creating a rolling validation window throughout the 2024-2025 season (Fig. 3a), with data splits elaborated in Supplementary Table S8. The weekly performance of our resulting 'Google Retrospective' model is detailed in the time-series leaderboard (Fig. 3b), which visualizes our model's performance advantage relative to the CovidHub-ensemble and other top-performing teams. The temporal variation of WIS for each of the separate validation splits is shown for the best replicate (Extended Data Fig. 5) and across replicates (Supplementary Fig. S6). A direct jurisdiction-level comparison confirms our model achieved a lower (better) WIS in a majority of states (Fig. 3c,d). Supplementary Figs. S7 and S8 compare prediction curves for representative locations between different models.

Overall, our model achieved the highest performance with an average WIS of 26, outperforming the official CovidHub Ensemble's average WIS of 29. A representative tree and breakthrough plot is shown in Extended Data Fig. 6.

Beyond this retrospective performance, we investigated ERA's ability to explore the solution space more broadly by replicating, recombining, and generating entirely new forecasting strategies (Fig. 3e). First, we tested its ability to replicate existing methods from other teams using only their brief public descriptions from the CovidHub (Supplementary Tables S9 and S10). Our tree-search-based implementations ('Base Method (TS)') not only adhered to the provided instructions (Supplementary Table S11) but also exceeded the performance of the original submissions in six of the eight cases tested; the two models that performed worse (replicating `JHU_CSSE-CSSE_Ensemble` and `OHT_JHU-nbxd`) did not use external data present for the original method implementations. Next, we explored whether solutions could be improved through recombination. For this experiment, we prompted an LLM to analyze the core principles of two different parent models and then used its synthesis to instruct ERA to generate a novel hybrid strategy combining their respective strengths. Of 26 generated hybrid models ('Recombination (TS)'), 11 achieved a WIS score superior to both of their parent models (Extended Data Fig. 7). We manually verified that output code for all recombination experiments contained relevant aspects from both parent codes (Supplementary Table S11). Finally, we used Gemini Deep Research[34] and AI co-scientist[35] to generate novel forecasting ideas which were then implemented using ERA. In total, this systematic exploration yielded 14 distinct strategies that outperformed the official CovidHub-ensemble: 10 from recombination, two from Deep Research, one from AI co-scientist, and one of our replicated baselines. Cosine similarities between embeddings for each generated code show clustering between different methods (Supplementary Fig. S9). An extended performance plot including the 9 TS models and 3 other submissions that did not outperform the baseline, alongside their corresponding validation performance, is provided in Supplementary Figs. S10 and S11. Examples of prediction curves are shown in Supplementary Fig. S12.

A deeper analysis of these 14 top-performing strategies reveals key patterns in how ERA achieves superior performance. The recombination models, which constitute the majority of the winners, highlight a clear pattern of synergistic hybridization. Two base models appear most frequently in these successful hybrids: the simple, climatology-based `CMU-climate_baseline` and the statistical autoregressive model `UMass-ar6_pooled`. This suggests ERA consistently discovers that the most effective strategies are built upon a robust foundation of historical averages and recent trends, which are then enhanced by more complex methods. Indeed, the most successful recombinations consistently fused different modeling paradigms: for instance, pairing the epidemiological `CEPH-Rtrend_covid` model with the statistical `UMass-ar6_pooled` model created a hybrid anchored in the theory of disease spread yet highly responsive to recent data trends, while pairing the powerful machine learning `UMass-gbqr` model with the stable `CMU-climate_baseline` provided a robust seasonal foundation that allowed the ML model to safely focus on learning short-term deviations. Finally, we verified the importance of using ERA, rather than just taking the best of 1000 solution attempts, for both epidemiological modeling and scRNA-seq batch integration (Table 1).

Time Series Forecasting: GIFT-Eval

We next evaluated ERA using General Time Series Forecasting Model Evaluation (GIFT-Eval)[19], a standard benchmark for time series forecasting. GIFT-Eval includes 28 datasets across seven diverse domains, with data frequencies ranging from seconds to years. It receives ∼4 new submissions per month that span diverse methodologies including black-box deep learning methods and foundation models. Submissions are scored on official train/validation/test splits using a normalized Mean Absolute Scaled Error (MASE) metric, calculated relative to a seasonal naive baseline.

We applied ERA in two phases. We began with a `per-dataset solution` in which ERA discovers an independent solution for each dataset. The second `unified solution` created a single general-purpose forecasting model using only basic libraries by hill-climbing against the average score for the entire GIFT-Eval.

Per-dataset solution Here we allowed ERA to use a full suite of Python libraries, including `scikit-learn`, `statsmodels`, and `xgboost`. ERA outperformed the entire May 18, 2025 leaderboard, which included foundation models, deep learning models, and standard time series methods (Supplementary Table S12). The discovered solutions showed strong convergence towards `gradient boosting` and `ensemble/decomposition` models (Supplementary Fig. S13).

Unified solution We wondered whether the code mutation system could create a unified, general-purpose forecasting library from scratch, by hill climbing with a *single* code on the average MASE on the entire GIFT-Eval dataset. To manage the benchmark's diversity, we allowed the library to have an adaptive configuration system, whereby it could generate up to 8 preset hyperparameter configurations to adapt to the diversity of datasets, with a validation step selecting the best performing configuration for each dataset. As the search progressed, date and trend-related features often led to performance breakthroughs leading to a model

that sequentially forecasts and subtracts individual time series components, including a base level, trend, seasonality, datetime-based features, and a final residual correction (Extended Data Fig. 8). The configurations (Supplementary Table S13) include date-specific features, including one that featurizes holidays in a specific set of countries (['US', 'DE', 'CN', 'GB', 'CA', 'AU']) . The resulting `unified solution` was highly competitive on the May 2025 leaderboard (Supplementary Table S12).

Other Problems: Geospatial Analysis, Neuroscience and Numerical Analysis

We also evaluated ERA on problems from three additional distinct domains: geospatial semantic segmentation, vertebrate neural activity forecasting, and numerical analysis (Supplementary Notes). ERA achieved expert-level performance in each case (Supplementary Figs. S14–S20, Supplementary Table S14).

Discussion

Our work introduces ERA, an AI-based system that drives a Tree Search (TS) with a Large Language Model (LLM) to systematically create and improve software for scientific tasks. By defining the problem of creating scientific software as a search for a program whose output maximizes a quality score, we convert software creation into a "scorable task", producing empirical software. ERA is novel in its LLM-driven rewriting approach, which allows for the flexible integration of domain knowledge and external research ideas. The ability of frontier LLMs to closely follow instructions enables efficient exploration of research ideas. ERA builds upon ideas from several distinct but related areas of research: Genetic Programming, Generative Programming, the application of LLMs to code, Automated Machine Learning (AutoML), Combining LLMs and Search, and agents for scientific discovery.

Genetic Programming — Genetic Programming (GP) provides a foundation to our work. In GP, a population of programs is iteratively improved using evolutionary principles like selection, crossover, and mutation. The fitness of each program is determined by a "fitness function," which is directly analogous to our "quality score"[36]. While GP has been successful, it traditionally relies on random mutations and structured recombination of code fragments (e.g., swapping sub-trees in an abstract syntax tree). Prior to the widespread adoption of large language models, GP systems successfully incorporated human expertise by restricting the search space using formal grammars or leveraging structural code metrics to bias the evolutionary search[37]. A key difference in ERA is the use of an LLM to perform intelligent, semantic-aware "mutations" by rewriting the code, which can produce more complex and meaningful variations than the random changes typical in GP.

Generative Programming — ERA can be viewed as a modern AI-driven realization of traditional generative programming, where, a developer creates a program generator (using techniques like templates, domain-specific languages[38], or metaprogramming) that produces tailored source code for a family of related problems[39]. In contrast, we use an LLM guided by a tree search as the generative engine. This approach offers greater flexibility, allowing

ERA to synthesize novel programs by exploring a vast solution space and integrating diverse domain knowledge in ways not easily achievable with more template-based methods.

LLMs for Code Generation — The advent of large language models pre-trained on vast code corpora has revolutionized code generation. Systems like AlphaCode[40] and OpenAI Codex[41] have demonstrated the ability to generate correct and complex code from natural language descriptions. These systems are typically used for "one-shot" generation from a prompt. Our approach differs by using the LLM in an iterative refinement loop. Instead of generating code from scratch, our LLM rewrites existing software candidates, guided by a search algorithm (TS) that uses the quality score as a signal.

AutoML — AutoML systems aim to automate the process of building machine learning pipelines by searching for optimal model architectures and hyperparameters. The goal is to maximize a performance metric (e.g., accuracy, F1-score) on a validation dataset[42], which fits our definition of a scorable task. While AutoML focuses specifically on finding the best model within a fixed set of ML frameworks, ERA is more general. It can rewrite any software, including pre-processing steps, complex simulations, mathematical heuristics, and other areas that fall outside the typical scope of AutoML.

Combining LLMs and Search — The most closely related work involves combining LLMs with search algorithms to overcome the limitations of one-shot generation. A rapidly growing body of literature formulates automated algorithm design as an Evolutionary Algorithm (EA)[43,44]. FunSearch uses an LLM to search for new mathematical discoveries by pairing a LLM suggesting code improvements with an automated evaluator[15]. Very recently, agentic frameworks like AlphaEvolve[14] have expanded these evolutionary coding loops towards needle-in-the-haystack discovery problems, similar in spirit to the tree search algorithms explored here, albeit without idea exploration required for scientific problem solving.

Agents for science problems — This sub-field has seen remarkable performance from highly specialized systems, with agents focusing on problems ranging from data science[45] to computational biology[46]. Instead of specializing in one domain, ERA demonstrates a general capability for empirical software optimization, achieving expert-level performance on public leaderboards and in academic literature across multiple fields.

To our knowledge, this is the first work demonstrating a system that beats human performance on a wide range of relevant and well-studied scientific problems. Across different fields, by prompting with the method description from cutting edge papers, ERA not only produces code that follows the methods of the paper but often has superior results.

We would like to emphasize the critical distinction between optimizing empirical predictive models and performing genuine scientific discovery, the latter of which requires reasoning about underlying theories, causal mechanisms, and mathematical frameworks. While the core problems evaluated in this paper primarily emphasize advanced empirical software engineering to allow for rigorous, automated scoring, our underlying system goes well beyond this, towards scientific discovery (Supplementary Notes).

The ability of LLM-based systems to autonomously produce expert-level empirical software carries broader safety risks, particularly when applied to domains that could directly impact human well-being. By automating complex engineering workflows, our system significantly lowers the technical expertise required to execute sophisticated computational tasks. While this democratization accelerates beneficial scientific discovery, it concurrently lowers the barrier to entry for deploying advanced models in sensitive or potentially dangerous domains.

This phenomenon represents a broader, systemic risk associated with the combination of large-scale inference-time compute and exponentially increasing foundation model quality.

To summarize, ERA combines a code mutation system based on Tree Search[2] with the ability to integrate complex research ideas. Such research ideas could come from the published literature, from research agents (e.g.[34,35]) or from combining previous ideas and solutions that the LLM has found itself. Because ERA creates code that can follow a specific idea, it can search over externally-supplied research ideas. ERA created 40 methods that beat the best known method for scRNA-seq batch integration and 14 methods that outperformed the CDC ensemble for epidemiological prediction. Additionally, ERA achieved expert-level performance on geospatial reasoning, neural activity prediction and algorithms for computational mathematics and a novel rule-based strategy for time series prediction.

Trial and error is essential to scientific progress, both for humans and for the automated approaches we outline here. ERA generates expert-level solutions extraordinarily quickly, reducing exploration of a set of ideas from weeks or months, to hours or days. Accelerating research in this way has profound consequences for scientific advancement. Based on this work, we believe that progress in scientific fields where solutions can be scored by machines is on the precipice of a significant acceleration.

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Figure Legends

Fig. 1 | Schematic and performance of ERA. **a**, Schematic of ERA algorithm. A scorable task, together with research ideas proposing methods to solve the task, are fed to an LLM, which produces code to evaluate the scorable task in a sandbox. This is then embedded within a tree search algorithm, whereby new nodes are chosen balancing exploitation and exploration, sampling from the LLM (Methods). **b**, Performance of code generation methods on Kaggle Playground benchmark. Results report the average public leaderboard percentile performance over 16 tasks. Methods based on ERA are listed in bold. TS, ERA-based tree search. BDT, boosted decision tree. Error bars indicate standard deviation of performance between different competitions in the benchmark. **c**, The system automates empirical software development by iteratively optimizing predictive models and computational algorithms based on research ideas and a defined quality metric. We use LLM summaries of scientific papers or AI assisted literature as part of the prompt, and also recombine successful implementations of ideas to create more powerful methods.

Fig. 2 | Performance of ERA on scRNA-seq batch integration. a, Schematic of the batch integration task, in which disparate datasets (teal and red) are processed to remove batch effects in the data while retaining biological variability. **b**, Performance of tree search (method names bolded and suffixed by "(TS)") compared to the analogous published method on the OpenProblems benchmark v2.0.0[30]. "Perfect embedding by celltype with jitter" is a positive control method that represents the best possible performance and "Shuffle integration by batch" is a negative control that does not perform any batch integration. Overall score is the mean over all datasets and metrics. Each Datasets column shows the mean of all metrics computed over that dataset. Each Metrics column shows the mean of that metric computed over all datasets. Metrics were assigned a value of 0 if they could not be computed or if their performance was worse than the lowest negative control; these are displayed as empty. **c**, Performance improvements annotated with code innovation for the top-performing batch balanced k-nearest neighbors (BBKNN) implementation. ComBat-based embedding generation was introduced in implementation attempt 429. **d**, Overall score for OpenProblems benchmark v2.0.0[30] non-control methods, ERA with and without recombination of ideas, Gemini Deep Research[34], and ERA with AI co-scientist[35]. Y-axis lower bound is the overall score of the "Shuffle integration by batch" negative control method. Seven recombination, five base methods, and two AI co-scientist methods that do not match its performance are omitted. * indicates the method is a recombination, even if not explicitly prompted for recombination. TS, ERA-based tree search; fastMNN, batchelor fastMNN; mnnCorrect, batchelor mnnCorrect.

Fig. 3 | Performance of ERA on COVID-19 forecasting. a, Rolling validation window used for the forecasting experiments. Each search's output is validated internally on a preceding block of time (blue), and the resulting model is then used to make predictions for its corresponding forecasting period (orange). Training data includes all dates on or after 2020-08-08 and prior to the validation set. **b,** Time-series leaderboard showing weekly forecasting performance (Average WIS) for participating teams and our 'Google Retrospective' model, ordered by average WIS. Scores are aggregated across all 52 jurisdictions and four forecast horizons. The number within each cell is the model's absolute Average WIS for that week. The cell's background color visualizes the performance relative to the CovidHub-ensemble, with blue indicating a lower (better) WIS and red indicating a higher (worse) WIS. **c,** Direct jurisdiction-level comparison of forecasting error (Average WIS) between our model and the 'CovidHub-ensemble', demonstrating our model's superior performance in a majority of locations. **d,** Geographic distribution of our model's forecasting error (Average WIS), aggregated over the entire 2024/25 COVID-19 season. Lower error values (lighter colors) indicate better performance. **e,** Comparison of aggregate forecasting performance for various modeling strategies. This includes baseline models from the CovidHub competition, our retrospective model, our replications of submitted models, novel hybrid models generated through recombination, deep research[34] and AI co-scientist[35]. 14 strategies (10 recombination; two Deep Research; one AI co-scientist and one replicated baseline) outperform the official CovidHub-ensemble for the 3-week (3 reference dates $\times$ 4 time horizons $\times$ 52 jurisdictions) evaluation period. Models that perform worse than CovidHub-baseline are not shown.

Tables

Table 1 | Performance comparison of ERA and Best-of-N on the scRNA-seq batch integration and epidemiological modeling tasks. Performance is measured on the validation set for each task. Bold entries indicate better performance. BoN, best-of-N=1000.

Model	Method	Batch integration (higher better)	Epidemiology (lower better)
Gemini 2.5 Flash	BoN	0.6306	106.55
	ERA	**0.6552**	**93.07**
Mistral Medium	BoN	0.6129	95.73
	ERA	**0.6332**	**87.98**
Claude Sonnet 4.6	BoN	0.6502	85.03
	ERA	**0.6575**	**84.56**
GPT-5	BoN	**0.6740**	78.04
	ERA	0.6671	**74.55**
Gemini 3.1 Pro	BoN	0.6461	92.39
	ERA	**0.6641**	**72.70**

Methods

Code mutation system

We prompt an LLM providing a description, the evaluation metric and the relevant data. The LLM produces Python code, which is then executed and scored on a sandbox. Searching over strategies dramatically increases performance: The system uses the score together with output logs and other information to hill climb towards a better score. We used a tree search (TS) strategy with an upper confidence bound (UCB) inspired by AlphaZero[47]. A critical difference from AlphaZero is that our problems don't allow exhaustive enumeration of all possible children of a node, so every node is a candidate for expansion. We therefore modify the UCB algorithm to count visits and compute mean values using the tree. However, when sampling a node to expand, we sample directly from the whole set instead of recursing from the root like AlphaZero. This makes our method closer to Flat UCB[48] than any MCTS variant like AlphaZero.

We also note that the algorithm differs from traditional TS, in that the scoring of the nodes do not involve random rollouts (e.g. of a game) to estimate the value of a node. Yet there is still randomness for scoring each node, caused by the sampling of the LLM itself, which produces a distribution of different codes (scores) for each fixed prompt.

We use a PUCT tree search algorithm to explore the space of code[2]. The PUCT (Predictor + Upper Confidence bound applied to Trees) algorithm is described in Algorithm 1. For tree T, and executed candidate u, we define the flat prior $P_T(u) = \frac{1}{|T|}$. To make it easier to tune the exploration constant c_{puct} across tasks, we convert task-specific scores TaskScore(u) to rank scores RankScore$_T(u)$ in the PUCT formula. We define RankScore$_T(u) = \frac{\text{Rank}_T(u)-1}{|T|-1}$, when $|T| > 1$, and 1 otherwise, where Rank$_T(u)$ gives ascending-order ranks to the candidates.

To select the next node for expansion, the algorithm balances exploitation of high-scoring solutions with exploration of the search space by computing a PUCT-inspired (Polynomial Upper Confidence Trees) acquisition score for each node i:

$$\text{PUCT}_i = r_i + c_{\text{puct}} \cdot E(i) \tag{1}$$

where c_{puct} is the exploration constant and $E(i)$ is an exploration term based on the node's visit count relative to the total number of visits across the entire tree. We tuned c_{puct} on the Kaggle benchmark to maximize performance, finding $c_{\text{puct}} = 1$ works well. The node with the globally maximum PUCT$_i$ score is selected. The language model then generates a single novel child solution conditioned on this parent's code and score. The new code is executed, assigned an empirical score, and appended to the search tree, followed by a backpropagation of the visit count to its ancestors. This global, flat-tree structure allows the system to seamlessly backtrack and branch from *any* historical node if the current optimization path plateaus.

We emphasize that in our algorithm mutations occur at the *code level* – ideas are part of prompts that produce code, that is then scored and iterated upon. With increasing nodes in the tree, the score saturates after 300-1000 nodes (See Breakthrough plots in Supplementary Figures). We search over ideas by combining different tree searches with different prompts together.

Finally, the implementation we outline here was chosen for optimal performance on the Kaggle Benchmark. Many alternatives were explored, including different agentic configurations

and the simple implementation outlined here had the highest overall performance across the benchmark. All experiments in this paper were carried out with Gemini 2.5 Flash, with the improvement with Gemini 2.5 Pro modest.

Algorithm 1 UCB tree search (PUCT)

Input: GenerateAndExecute(), TaskScore() to define rank scores $\text{RankScore}_T(u)$, exploration constant c_{puct}, and a root node r.

$T \leftarrow \{r\}$ ▷ Initialize the tree with a root node.
$V(r) \leftarrow 1$
for all iterations **do**
  $N_{total} \leftarrow \sum_{u \in T} V(u)$ ▷ Get total visits across all nodes
  Select $u^* \leftarrow \text{argmax}_{u \in T} \left(\text{RankScore}_T(u) + c_{puct} P_T(u) \frac{\sqrt{N_{total}}}{1+V(u)} \right)$ ▷ Select node with highest PUCT score
  $u_c \leftarrow \text{GenerateAndExecute}(u^*)$ ▷ Expand the selected node and Execute
  $T \leftarrow T \cup \{u_c\}$
  $V(u_c) \leftarrow 1$
  for all ancestors u_a of u_c (excluding u_c) **do** ▷ Backpropagate results
    $V(u_a) \leftarrow V(u_a) + 1$
  end for
end for
return $\text{argmax}_{u \in T} \text{TaskScore}(u)$ ▷ Best solution found

It is useful to explicitly contrast this algorithm with standard heuristic search algorithms. Our approach does not rely on beam search (which aggressively prunes unselected candidates at each depth layer) nor a $(1 + \lambda)$ evolution strategy (which maintains only the most recent generation and discards historical states).

Similarly, our system differs from emergent efforts to incorporate LLMs into GPs [43,44,49–51].

More recently, agentic AutoML frameworks encompass automated feature engineering, meta-learning, dynamic pipeline construction, and multi-objective optimization [52], including agentic systems [53,54], which have shown more impressive performance [55].

Adding research ideas to the code mutation system

When an expert solves difficult scientific problems, they often search for prior work for ideas. Prior work could be sourced from highly cited papers, specialized textbooks, or search engines. The search for prior work can also be powered by LLMs [34,35,56–60].

We emulate the expert behavior by injecting instructions for carrying out research ideas into the prompt of our code mutation system (Fig. 1). We applied the research instruction injection for scRNA-seq batch integration, COVID prediction, segmenting remote sensing images, and whole-brain neural activity prediction. While the most successful outcomes used top methods from the literature, we also used two LLM driven search strategies: Deep Research from Gemini 2.5 Flash [34] and AI co-scientist [35].

For running these searches, we provided the tools with background information from the main problem description, and instructed the models to create distinct ideas (Supplementary

Table S15). After manually filtering proposals and removing one proposed scRNA-seq batch integration method, we prompted Gemini to format the ideas into a structure consistent with our baseline method descriptions (Supplementary Table S16). Finally, we ran ERA on these ideas to create empirical codes that could be scored.

ERA versus best-of-N

We use $N = 128$ ERA search nodes and compare the performance of Gemini 2.5 Flash, Mistral Medium, Claude Sonnet 4.6, GPT-5, and Gemini 3.1 Pro on the scRNA-seq batch integration task and an epidemiological flu forecasting task. This explores a range of models that were contemporaneous with the core model used in this paper (Gemini 2.5 Flash) while also using Gemini 3.1 Pro as a state-of-the-art model at the time of paper publication. The scores use the same scoring rules as in all other experiments: for the epidemiological forecasting task it is the Weighted Interval Score evaluated over a rolling evaluation window, while for scRNA-seq batch integration it is the average of metrics measuring preservation of biological information while eliminating batch effects, both measured on the validation set. Note that significant improvements in batch integration happen with much smaller change in validation scores than in epidemiological forecasting. We present here the best performance over 3 different experiments (with both Best-of-N and ERA). Typical variation in performance is of order 0.01 for batch integration and $O(1)$ for epidemiological forecasting, with some model dependence. For example, GPT-5 has much more experiment-to-experiment variability than other models.

We note that ERA explores the solution space more efficiently than Best-of-N, and outperforms Best-of-N for all models and problems, with the exception of GPT-5's performance on batch integration. On the batch integration problem, GPT-5's one shot performance is sufficiently good that this distinction isn't important. Indeed we expect that as frontier models improve, tasks that were once difficult will saturate. Tree search is useful for pushing model performance on difficult tasks.

Recombination experiments

For both scRNA-seq batch integration problem and COVID-19 forecasting, we combined ideas from methods already generated using tree search. For the scRNA-seq batch integration problem, we used the first versions of our 11 baseline methods. For the COVID-19 prediction problem, we used the eight replications of models submitted to CovidHub. We first took the top-performing node from each tree search run seeded with one of these methods, based on its score on the validation set (for COVID-19 prediction, this included six weeks of reference dates from 2025-02-22 to 2025-03-29). Then, for every pair of these methods, we prompted Gemini 2.5 Flash to compare the two methods and explain the core technical similarities and differences between the two parent models using a consistent prompt (Supplementary Table S7). The explanatory response was then added to the the prompt, along with a statement instructing tree search to recombine the ideas by combining the best parts of both approaches (Supplementary Table S17). Subsequently, we ran ERA to generate new hybrid strategies. This process yielded 55 recombined methods for the scRNA-seq batch integration

problem, and 28 for the COVID-19 prediction problem (evaluated on the three-week holdout set 2025-04-05 to 2025-04-19, see Fig. 3e).

Our method for recombination differs from previous efforts where the LLM acts as a mutation and crossover operator[49–51,61–64]. Our approach to recombining expert ideas is conceptually related to evolutionary methods such as RHEA, with the comparative advantage that our tree search recombines expert ideas directly in conceptual space via LLM prompts rather than solely in distilled model space[65].

Gemini embeddings

For each tree search implementation, we input the code snippets to the Gemini text embedding model[66], and the resulting 3,072-dimensional output vectors served as the semantic representations of their respective implementations.

scRNA-seq batch integration

For all scRNA-seq experiments, we ran tree search with 500 nodes. Each experiment took roughly seven hours to execute on our infrastructure.

Dataset We sourced a dataset from CZ CELLxGENE Discover[25] to use for hill climbing with tree search. To identify datasets distinct from the six OpenProblems.bio test datasets but that have similar characteristics, we filtered to datasets that contain only healthy human cells, with primary cell count $\geq$ 2,000, at least 10 unique cell types, at least seven unique donor ids (i.e. number of batches), and contain at least two unique assays that are also present in the OpenProblems.bio datasets. This filtering process identified 22 candidate datasets. After manually investigating the candidate datasets, we selected the dataset `364bd0c7-f7fd-48ed-99c1-ae26872b1042` version `ffdaa1f0-b1d1-4135-8774-9fed7bf039ba`[17].

Within the selected dataset, we applied quality control metrics and data processing steps identical to the processing performed on the OpenProblems.bio datasets[67,68], yielding a processed dataset with normalized expression values, highly variable genes, principal components, and k-nearest neighbors all computed. For computational efficiency, we randomly selected two disjoint subsets of $N = 20,000$ cells each, attempting to match `(batch, cell type)` distributions of the entire processed dataset. The "train" dataset was used for model training and selection of the highest-performing node in a single tree search. The "validation" dataset was used to select the best tree search for methods in which we ran multiple replicates of the same algorithm (Supplementary Fig. S1).

Evaluating scRNA-seq Batch Integration on the OpenProblems.bio Benchmark We downloaded the OpenProblems `v2.0.0` input and solution data from `s3://openproblems-data/resources/task_batch_integration/datasets/cellxgene_census/` and raw performance metrics from `s3://openproblems-data/resources/task_batch_integration/results/run_2025-01-23_18-03-16/score_uns.yaml`. We computed control-scaled metric results identically to the published OpenProblems results. Briefly, for each `(dataset, metric)`, lower and upper bounds on raw scores are defined as the minimum and maximum values achieved

by the seven "control" methods. Raw values were linearly scaled between those extrema and clamped to be in $[0, 1]$. Overall score was computed as the arithmetic mean over all 78 measurements (13 metrics computed for each of 6 datasets) with NaN values replaced by 0 (i.e., failure to compute a metric causes it to be considered the worst possible score).

Replication of Existing Methods for Batch Integration The OpenProblems.bio benchmark profiles the performance of several state-of-the-art existing methods. As of July 11, 2025 there were 19 different methods. Three methods have implementations in both R and Python: `LIGER` and `pyliger`, `Harmony` and `Harmonypy`, and `batchelor mnnCorrect` and `mnnpy`. After grouping reimplementations of the same method, there are 16 separate research ideas. From this list, we excluded all six foundation model methods (`UCE`, `SCimilarity`, `scGPT (zero shot)`, `scGPT (fine-tuned)`, `Geneformer`, and `scPRINT`) because they perform very poorly on the benchmark and use a much larger training set. For example, only a single foundation model (`UCE`) performs better than the negative control of "No integration" which simply performs PCA on the dataset. We further excluded `scANVI`, which is a modification of `scVI` that is trained using cell type information. Since cell type information is used to define the metrics, this represents data leakage and consequently we consider `scANVI` a control method. This resulted in nine existing different research methods to optimize with tree search.

For each of the nine existing methods, we obtained the manuscript PDF corresponding to the method. To obtain a short method description from the manuscript, we used Gemini 2.5 Pro Thinking to summarize the paper (prompt in Supplementary Table S18, example output in Supplementary Table S19). For `batchelor fastMNN`, which is a faster implementation of `batchelor mnnCorrect`, there is no separate publication and thus we provided the paper PDF of `batchelor mnnCorrect` as well as the docstring corresponding to `batchelor fastMNN` from https://rdrr.io/github/LTLA/batchelor/man/fastMNN.html (Details section) with a slightly adjusted prompt. Finally, the method summary is added to the tree search prompt, and is used to come up with better code solutions given the method summary.

For each of the nine methods, we ran three replicates of tree search. For Fig. 2, we selected the replicate that had the best performance based on the validation set score. We show the performance of all replicates in Extended Data Fig. 1. Code for the best performing method is in Supplementary Table S6.

Hyperparameters To determine optimal hyperparameters for each base method, we employed Optuna, an automated hyperparameter optimization framework[69]. Search spaces were defined across integer, float, and categorical parameter types by experts. The optimization process ran for a total of five times the number of parameters. In each trial, a model was trained using a sampled parameter set and evaluated based on a performance metric that Optuna's Tree-structured Parzen Estimator (TPE) sampler aimed to maximize. All hyperparameter optimization was conducted solely on the training dataset. The best identified hyperparameter set was then utilized to train the final base methods and evaluate them on the held-out OpenProblems dataset.

COVID-19 prediction

Dataset Our primary data source was historical confirmed COVID-19 hospital admissions, which corresponds to the target variable specified by CovidHub. These data are published weekly by the CDC within the National Healthcare Safety Network (NHSN) Hospital Respiratory Data (HRD) dataset[70]. Preprocessing was kept minimal–missing values in the dataset were replaced by zeros to enable tree search to find executable code with the criterion score (WIS). The only additional data source used to augment the target for our model was static jurisdiction-specific population values from the CovidHub GitHub Repository[18]. For comparing model performance in Fig. 3c, we use all of the models submitted to Forecast Hub which make predictions at a state by state level and have forecasts for at least 75 percent of the season and time horizons. We ran tree search with 2000 nodes for each reported run. We note that we use available as of 2025-05-01 for the entire retrospective season, thus ignoring potential differences in data available at the date of forecast.

Replication of existing COVID-19 prediction models We selected eight models for replication from those that had submitted to CovidHub based on the following inclusion criteria: (1) The method must be reproducible solely using historical COVID-19 hospitalization data, without reliance on external predictor variables, (2) The model submission must include predictions across all specified time horizons, and (3) Model submissions must be available for over three months (12 weeks) to enable meaningful comparison. Three models were excluded for failing these criteria: two were ensembles of external forecasts, and one relied entirely on additional data. An additional five models were excluded because they did not provide predictions for all forecast horizons. These five models originated from the same forecasting team. As all our analysis involves aggregating model performance across horizons, we have excluded these five models from all comparisons. Overall this gave a selection of eight models for replication.

To instruct the search algorithm, we provided the method descriptions from the original authors' official submission metadata. For example, the metadata for the `UMASS-arc6-pooled` model states: "*AR(6) model after fourth root data transform. AR coefficients are shared across all locations. A separate variance parameter is estimated for each location.*" We integrated these concise descriptions directly into the tree search prompt as part of the model directions, transforming them into instructions by prepending 'Use a/an' (see Methods, Supplementary Table S10).

GIFT-Eval benchmark

We applied our tree search methodology to the General Time Series Forecasting Model Evaluation (GIFT-Eval) benchmark[19]. The search begins from a root node defined by an initial code template and proceeds via hill climbing, where new candidate solutions are generated and evaluated against the GIFT-Eval validation folds. At the end of a tree search, we evaluated the solution on the held-out test set using MASE point forecast as the scoring metric. Our results are based on a 5/18/2025 snapshot of the dataset, official leaderboard and scoring, all of which have been updated since. See Supplementary Table S12 for a complete snapshot of the leaderboard.

We remark that our snapshotting to a fixed date ensures stability. The GIFT-Eval protocols underwent significant structural changes after our experiments concluded, including major scoring/dataset corrections on July 24, 2025, and the introduction of an "Agentic" category on August 5, 2025. Later updates on August 25 redefined "Zero-shot" to account for widespread test data leakage in foundation models using the large pre-train dataset (Note that our method does not use the large pre-training data). To avoid unsound comparisons against baselines developed under these revised protocols, we deliberately chose not to submit to the live public leaderboard, restricting our comparison to the May 18th snapshot to ensure a valid, stable baseline. We also note that Gemini models driving our Tree Search have evolved significantly since our experiments.

We adhered to the benchmark's framework, utilizing the official dataset source from Hugging Face, its pre-defined training, validation, and test splits, as well as the scoring and evaluation code commonly used in the existing submission notebooks.

Per-dataset Solution We conducted separate tree searches for 92 of the 97 GIFT-Eval datasets, excluding the five largest due to computational constraints; for these, the naive baseline score was used in order to produce the aggregated leaderboard score. For each dataset, we used a search of 300 nodes, with the system permitted to use a broad suite of machine learning libraries, including `scikit-learn`, `XGBoost`, and `statsmodels`. Supplementary Fig. S13 shows an analysis of the types of models used across the 92 different solutions.

Unified Solution Here, we created a single, unified forecasting library that could generalize across all 97 datasets. We used a tree search of over 1,000 nodes, guided by the geometric mean of the normalized MASE scores across all datasets, providing a single objective function to optimize. To force the model to reason from first principles, its access was restricted to basic libraries (`numpy`, `pandas`, and `holidays`).

The resulting solution consists of two components: a single forecasting library and a list of eight preset configurations. For each dataset, the best-performing configuration is identified on the validation set. This selected configuration is then used with the unified library to produce the final forecast on the test set, allowing the model to adapt its strategy without seeing test data.

The final solution was developed iteratively. An initial search yielded a base model with a MASE of 0.82. A key breakthrough occurred in a subsequent run when the search space was expanded to ten configurations and the system was advised to use the `holidays` library, which improved the MASE to 0.77 (Extended Data Fig. 8). A final 500 node refinement run pruned the configurations to an optimized set of eight, achieving the final MASE of 0.734.

The final solution sequentially models and removes fundamental components of the series, with the final forecast being the sum of the individual component forecasts. This approach allows the model to be highly configurable while systematically accounting for different sources of variation in the data. This process is outlined with the following steps:

1. **Preprocessing:** The input series first undergoes basic cleaning, including median imputation for any missing values. An optional log-transform (`log1p`) can be applied to stabilize variance in series with exponential growth patterns.

2. **Base/Level Component:** A base level is established using simple but robust methods like a seasonal naive forecast or a rolling median of recent data points. This component captures the basic magnitude of the series.

3. **Trend Component:** The residuals from the base component are then modeled to capture linear or polynomial trends. This step includes a `damping_factor` to prevent unrealistic long-term extrapolation by gradually flattening the trend.

4. **Seasonality Component:** The residuals from the trend component are analyzed to model cyclical patterns (e.g., weekly, yearly). The model identifies the cycle length and forecasts seasonality by averaging values at the same point in the cycle (e.g., the average value for all Mondays).

5. **Datetime and Holiday Features:** To capture special events and non-seasonal cycles, features are extracted from the timestamp (e.g., `dayofweek`, `is_holiday_flag`). The model calculates the median effect of each feature category from the remaining residuals and adds it to the forecast.

6. **Residual Correction:** As a final step, a correction is made by modeling the median of the most recent unexplained errors. This autoregressive-like step helps correct for short-term biases in the model. A `decay_factor` fades its impact over the forecast horizon.

To apply the unified solution to a new dataset, one would first split the historical data into training and validation sets. Using the library's adaptive configuration system, one can then find a suitable forecasting strategy by evaluating the eight preset configurations on the validation data to select the best-performing one. This provides a strong, data-driven starting point that can be used directly. For more specialized applications, one can also create a custom configuration, allowing for manual refinement of the model's components and making the library both powerful out-of-the-box and flexible enough for expert tuning.

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Code Availability

A reference implementation of ERA is available at https://github.com/google-research/era. The best candidate solutions generated for each of the six scientific problems in this paper are publicly available at https://google-research.github.io/era, along with a user interface enabling examination of the full tree search data for a representative run for each of the six scientific problems. The interface allows inspecting the solution progression and breakthrough plot as the tree search proceeds and highlights code diffs.

Author Contributions Statement

Code Mutation System (E.A., A.B., G.C., M.C., H.C., P.N., D.S., J.T., S.V., M.P., J.K., P.R., J.W., L. W., S.M. and M.P.B.) Single Cell RNA-seq Batch Integration (A.Be., C.Y.M., C.H., Y.Z., M.P.B.) COVID Forecasting (Z.S., S.M., M.P., M.C., M.P.B.) Geospatial Analysis (R.J., Je.C., Q.Z., M.P.B.) ZAPBench (B.P.W., J-M.L, Q.Z.) GIFT-Eval (J.G., M.P.B.) Integrals (A.K., R.K., M.P.B.) User Interfaces (E.A., G.C., P.N., A.K., M.K., M.P.B., J-M.L., D.L., J.K., C.C., S.E.) Graphical Design (G.J.) Program Management (M.A., E.B.) Leadership (K.C., J.M., Y.M., J.C.P., L.D., S.M., M.P.B.)

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Competing Interest Declaration

The authors affiliated with Google are employees of Google Inc. and hold Alphabet stock.

Additional Information

Supplementary information is available online.

Corresponding Authors: Michael P. Brenner (mbrenner@google.com) and Shibl Mourad (shibl@google.com).

Extended Data Figure Legends

Extended Data Fig. 1 | Relative performance of base methods and ERA replicates. a, Overall scores on the holdout OpenProblems datasets for all replicates of methods evaluated in Fig. 2. For tree search implementations, three replicates of the full process were performed. Dots indicate the overall score of the replicate on the holdout OpenProblems datasets. The bar shows the performance of the replicate with highest performance in the validation dataset (identical values to those shown in Fig. 2). The lowest performing tree search replicates for BBKNN, Scanorama, and TabVI only successfully computed 30, 57, and 45 of the 78 metrics, respectively. We note that failures due to out of memory or compute time issues were not explicitly selected against in our algorithm since all optimization was performed on datasets of only 20k cells. **b,** Average scores for each method when restricting to only `(method, dataset, metric)` combinations that have non-NaN values for the base method and all three tree search replicates. `No advice` and `TabVI` are absent since they have no base method comparator.

Extended Data Fig. 2 | Breakthrough plot and solution tree for the scRNA-seq batch integration task. *Top Figure* Breakthrough plot for the `BBKNN (TS)` tree search, showing the evolution of the maximum score as a function of the number of nodes. The green dots label places where the score abruptly increases due to an improvement in the code, and the label describes the change in the code that resulted in the score increase. *Bottom Figure* Structure of the tree for this same search. The color range consists of orange (lower scores) to green (higher scores) with the highest score denoted by a diamond node.

Extended Data Fig. 3 | Ablation analysis of the top-performing `BBKNN (TS)` method. The `BBKNN (TS)` method performed standard linear expression scaling to 10^4 total counts followed by `log1p` transformation. It then applied three additional transforms: "Standardize" called `sc.pp.scale` to further scale the data to mean 0 and unit variance, "ComBat+PCA" called `sc.pp.combat` followed by `sc.tl.pca` to generate the expression embedding, and "BBKNN" applied an implementation of batch-balanced k-nearest neighbors written by ERA. Bars here show the overall performance in the OpenProblems datasets for ablations that include one or more of these components. For each ablation that includes the "BBKNN" component, comparison of the written BBKNN implementation ("Tree search") and the `bbknn` package implementation ("Package") is shown. Black dots show individual performance of three replicates of each method.

Extended Data Fig. 4 | Comparison of tree search performance on base methods and their "recombination" over an intersection of successfully calculated metrics. We ran "recombination" experiments by seeding tree search with the top variants from two base method runs (see Methods). We compare the performance of two base methods and "recombination" on the OpenProblems test dataset for all 55 pairwise combinations of the 11 base methods. Since sometimes methods may fail getting a score for certain evaluation metrics due to errors like out of memory, we compare the performance on a subset of metrics that were successfully computed for all three methods. "n=X/78" on each subplot shows the number of successfully computed metrics, X, that we averaged over. For each subplot, we show the base methods on the left in light blue, and the recombination method on the right (labeled as "Recomb"), where a green bar means the recombination method outperforms both of its base methods, dark blue means the recombination method outperforms one of the base methods, and red means the recombination method does not outperform either of the base methods.

Extended Data Fig. 5 | Performance of the best retrospective COVID-19 hospitalization forecast replicates. This figure presents WIS by reference date for the single best-performing replicate of each validation window in our retrospective COVID-19 forecasting study. The best models are selected based on their performance on the validation dates. The plot shows how finding optimum models on a handful of validation dates (6 weeks) generalizes to the next two weeks of unseen reference dates.

Extended Data Fig. 6 | Breakthrough plot and solution tree for the COVID-19 prediction task. *Top Figure* Breakthrough plot for the retrospective COVID-19 prediction, showing the evolution of the maximum score as a function of the number of nodes. The green dots label places where the score abruptly increases due to an improvement in the code, and the label describes the change in the code that resulted in the score increase. *Bottom Figure* Structure of the tree for this same search. The color range consists of orange (lower scores) to green (higher scores) with the highest score denoted by a diamond node.

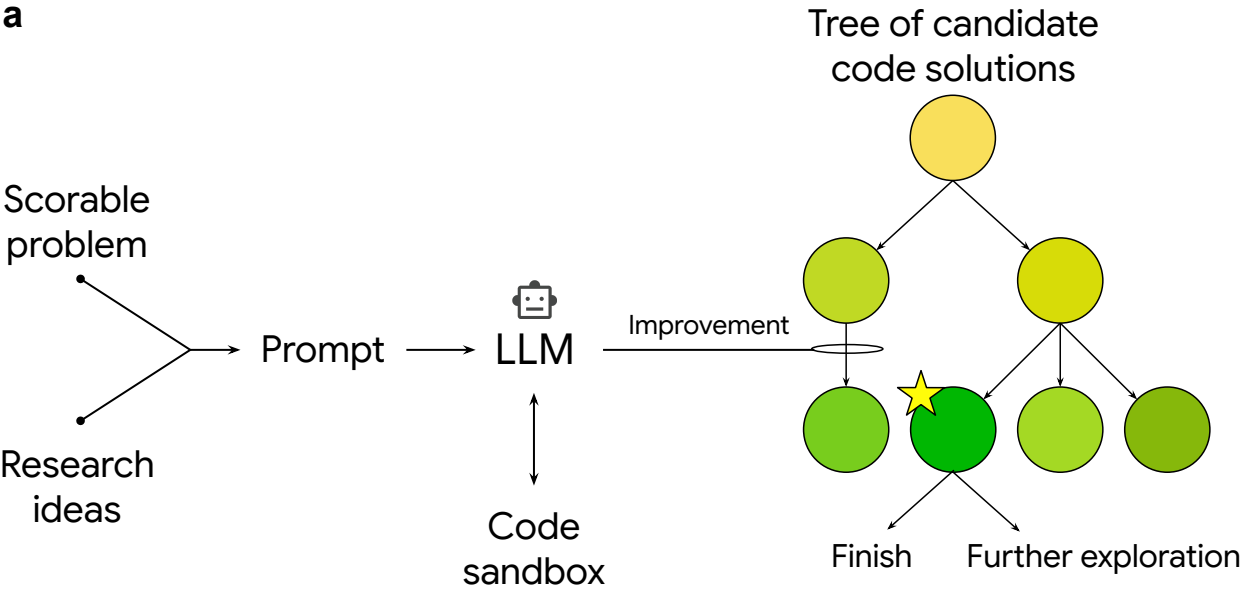
Extended Data Fig. 7 | Performance of recombination experiments for COVID-19 forecasting. This series of bar plots illustrates the average WIS achieved by various hybrid models (right bar, labeled "Recomb") compared to their constituent baseline models (left bars, typically light blue) from the CovidHub competition. Each subplot represents a recombination experiment, demonstrating the success of our system in synthesizing novel forecasting strategies. Green bars indicate that the recombination outperformed both parent models, dark blue indicates it outperformed one, and red indicates it outperformed neither. These results emphasize the search system's ability to combine the strengths of existing methodologies to achieve superior predictive performance.

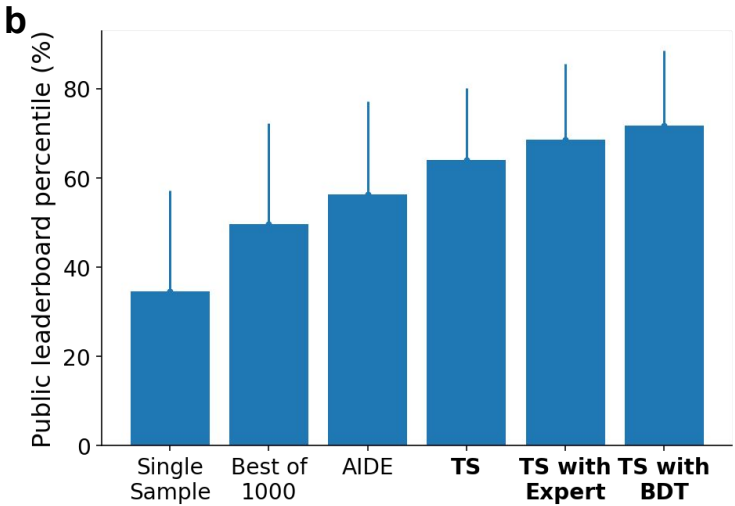
Extended Data Fig. 8 | Breakthrough plot and solution tree for the time series prediction task. *Top Figure* Breakthrough plot for the GIFT-Eval tree search, showing the evolution of the maximum score as a function of the number of nodes. The green dots label places where the score abruptly increases due to an improvement in the code, and the label describes the change in the code that resulted in the score increase. *Bottom Figure* Structure of the tree for this same search. The color range consists of orange (lower scores) to green (higher scores) with the highest score denoted by a diamond node.

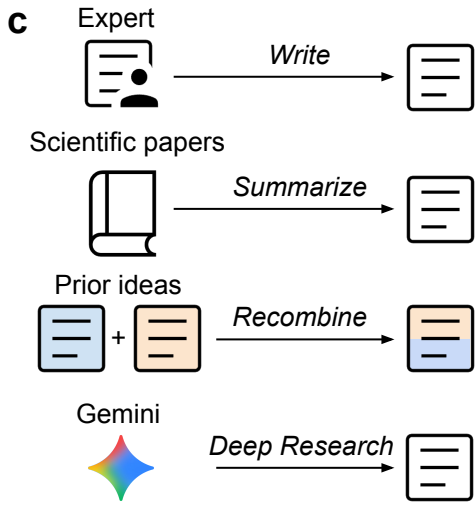
Extended Data Tables

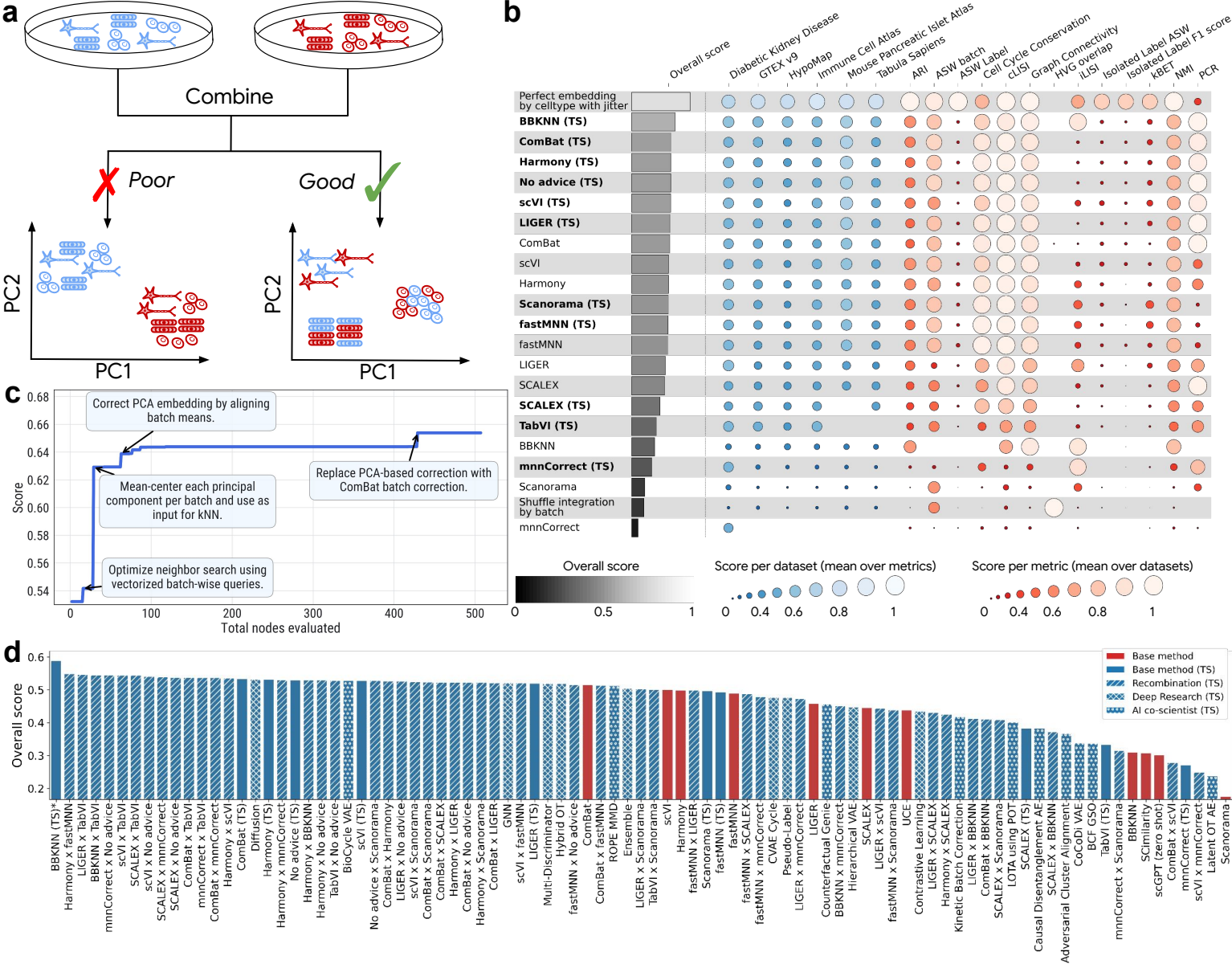
Extended Data Table 1 | Expert manual inspection of adherence of ERA implementation to scRNA-seq batch integration method.

Method	Replicate	Judgment	Notes
batchelor fastMNN	0	Follow	
batchelor fastMNN	1	Follow	
batchelor fastMNN	2	Follow	
batchelor mnnCorrect	0	Follow	
batchelor mnnCorrect	1	Follow	
batchelor mnnCorrect	2	Follow	
BBKNN	0	Follow	Adds distances between batches, performs spectral clustering on the graph. Does not compute connectivities.
BBKNN	1	Follow + Innovative	Standardize + ComBat + PCA for embedding. BBKNN implemented on that embedding.
BBKNN	2	Follow	Corrects the data, computes neighbors, final embedding is UMAP supposedly based on neighbors.
ComBat	0	Follow	
ComBat	1	Follow	
ComBat	2	Follow	
Harmony	0	Follow	Entropy-based diversity penalty.
Harmony	1	Follow	Linear diversity penalty.
Harmony	2	Follow	Linear diversity penalty.
LIGER	0	Follow	Uses `sklearn.NMF` with multiplicative update solver.
LIGER	1	Follow	Writes NMF function from scratch. Builds single global KNN graph rather than by batch.
LIGER	2	Does not follow	Uses ComBat + SVD.
No advice	0	Follow	Uses batch-specific mean+std for all genes to rescale. Then PCA.
No advice	1	Follow	ComBat + SVD
No advice	2	Follow	ComBat + PCA
SCALEX	0	Follow	Adds log_var clipping and weight normalization.
SCALEX	1	Follow	Learns batch embedding. Learns gamma and beta conditioned on batch index. Batch index not supplied to first layer of decoder.
SCALEX	2	Follow	Uses min_delta for robust early stopping. batch_index not supplied to the first layer of the decoder.
Scanorama	0	Follow	
Scanorama	1	Does not follow	Implements mnnpy via `sc.external.pp.mnn_correct`.
Scanorama	2	Follow	
scVI	0	Follow	Applies log1p scaling with ZINB loss. Fits global dispersion theta rather than batch-specific.
scVI	1	Follow	Applies optional log1p scaling with ZINB loss. Fits global dispersion theta rather than batch-specific.
scVI	2	Follow	Expression frequency exponentiated rather than softmaxed. Applies log1p scaling with ZINB loss. Fits global dispersion theta rather than batch-specific.
TabVI	0	Follow	
TabVI	1	Follow	
TabVI	2	Follow	


a
Tree of candidate code solutions
Scorable problem
Research ideas
Prompt
LLM
Improvement
Code sandbox
Finish
Further exploration


b
Public leaderboard percentile (%)
0
20
40
60
80
Single Sample
Best of 1000
AIDE
TS
TS with Expert
TS with BDT


c
Expert
Write
Scientific papers
Summarize
Prior ideas
Recombine
Gemini
Deep Research


a
Combine
Poor
Good
PC2
PC1
PC2
PC1
b
Overall score
Diabetic Kidney Disease
GTEX v9
HypoMap
Immune Cell Atlas
Mouse Pancreatic Islet Atlas
Tabula Sapiens
ARI
ASW batch
ASW Label
Cell Cycle Conservation
cLISI
Graph Connectivity
HVG overlap
iLISI
Isolated Label ASW
Isolated Label F1 score
kBET
NMI
PCR
Perfect embedding by celltype with jitter
BBKNN (TS)
ComBat (TS)
Harmony (TS)
No advice (TS)
scVI (TS)
LIGER (TS)
ComBat
scVI
Harmony
Scanorama (TS)
fastMNN (TS)
fastMNN
LIGER
SCALEX
SCALEX (TS)
TabVI (TS)
BBKNN
mnnCorrect (TS)
Scanorama
Shuffle integration by batch
mnnCorrect
Overall score
0
0.5
1
Score per dataset (mean over metrics)
0
0.4
0.6
0.8
1
Score per metric (mean over datasets)
0
0.4
0.6
0.8
1
c
Score
Total nodes evaluated
Correct PCA embedding by aligning batch means.
Mean-center each principal component per batch and use as input for kNN.
Optimize neighbor search using vectorized batch-wise queries.
Replace PCA-based correction with ComBat batch correction.
d
Overall score
Base method
Base method (TS)
Recombination (TS)
Deep Research (TS)
AI co-scientist (TS)
BBKNN (TS)*
Harmony x fastMNN
LIGER x TabVI
BBKNN x TabVI
mnnCorrect x No advice
scVI x TabVI
SCALEX x TabVI
scVI x No advice
SCALEX x mnnCorrect
SCALEX x No advice
ComBat x TabVI
mnnCorrect x TabVI
ComBat x mnnCorrect
Harmony x scVI
ComBat (TS)
Diffusion
Harmony (TS)
Harmony x mnnCorrect
No advice (TS)
Harmony x BBKNN
Harmony x No advice
TabVI x No advice
BioCycle VAE
scVI (TS)
No advice x Scanorama
ComBat x Harmony
LIGER x No advice
scVI x Scanorama
ComBat x Scanorama
ComBat x SCALEX
Harmony x LIGER
ComBat x No advice
Harmony x Scanorama
ComBat x LIGER
GNN
scVI x fastMNN
LIGER (TS)
Multi-Discriminator
Hybrid OT
fastMNN x No advice
ComBat
ComBat x fastMNN
ROPE MMD
Ensemble
LIGER x Scanorama
TabVI x Scanorama
scVI
Harmony
fastMNN x LIGER
Scanorama (TS)
fastMNN (TS)
fastMNN
fastMNN x SCALEX
fastMNN x mnnCorrect
CVAE Cycle
Pseudo-Label
LIGER x mnnCorrect
LIGER
Counterfactual Genie
BBKNN x mnnCorrect
Hierarchical VAE
SCALEX
LIGER x scVI
fastMNN x Scanorama
UCE
Contrastive Learning
LIGER x SCALEX
Harmony x SCALEX
Kinetic Batch Correction
LIGER x BBKNN
ComBat x BBKNN
SCALEX x Scanorama
LOTA using POT
SCALEX (TS)
Causal Disentanglement AE
SCALEX x BBKNN
Adversarial Cluster Alignment
CoCoDi VAE
BCF GSO
TabVI (TS)
mnnCorrect x Scanorama
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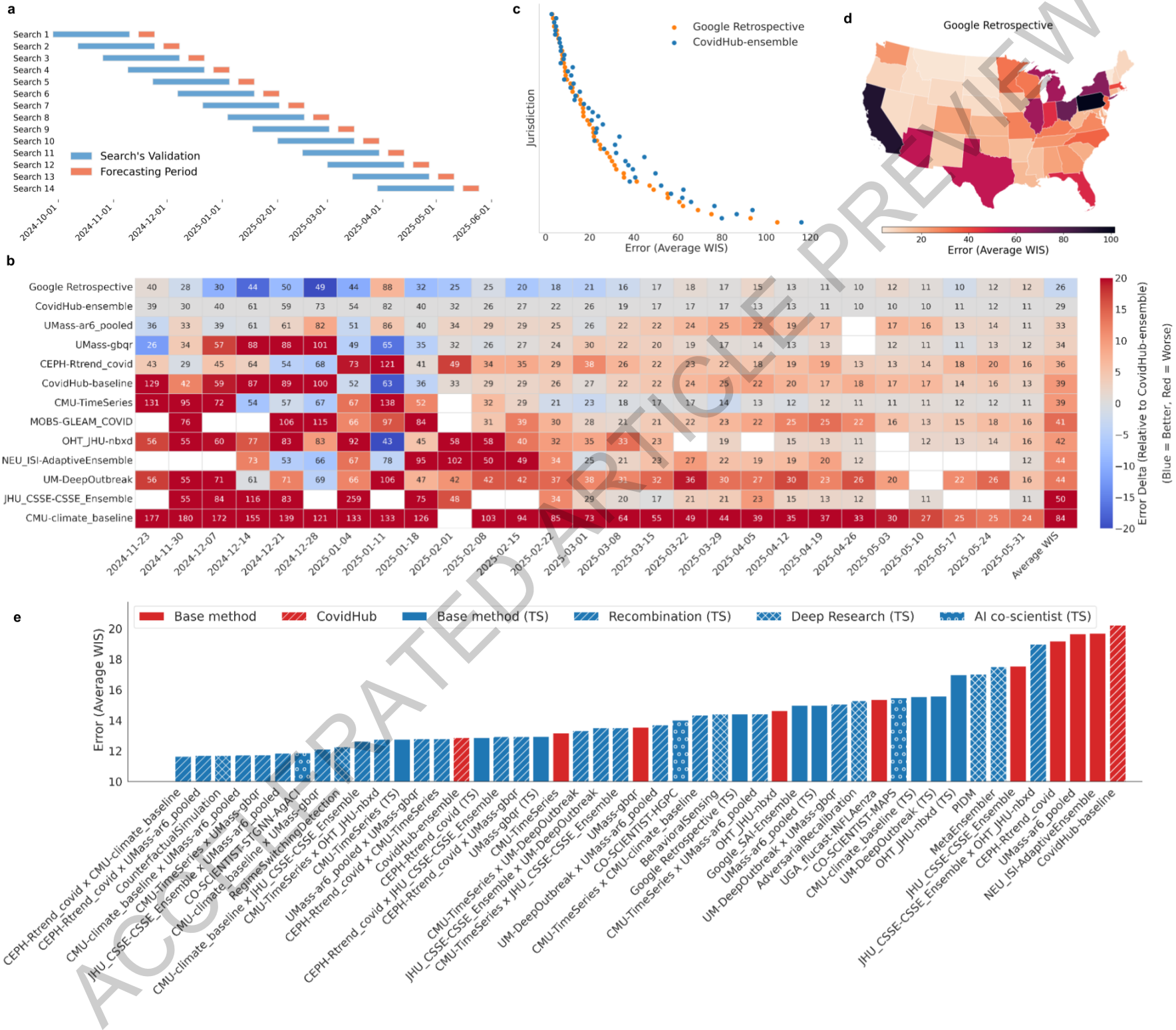
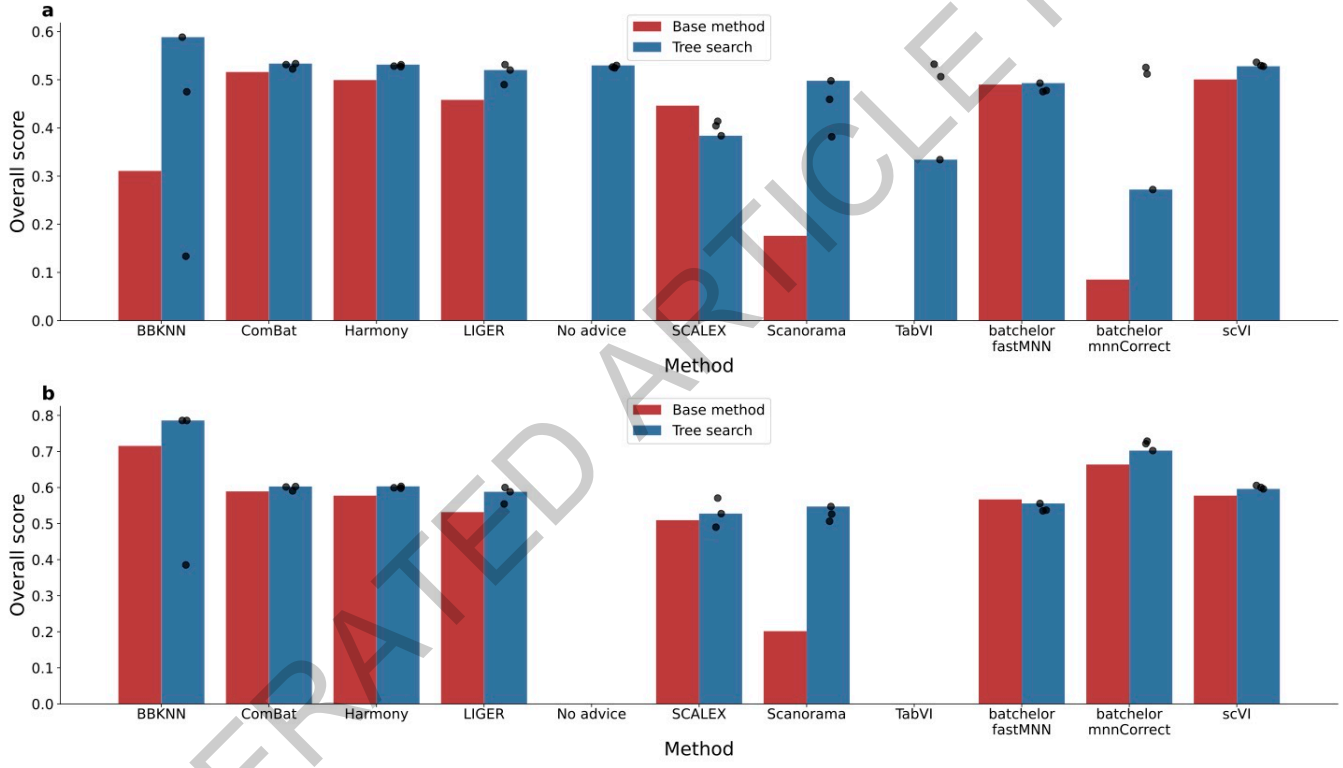
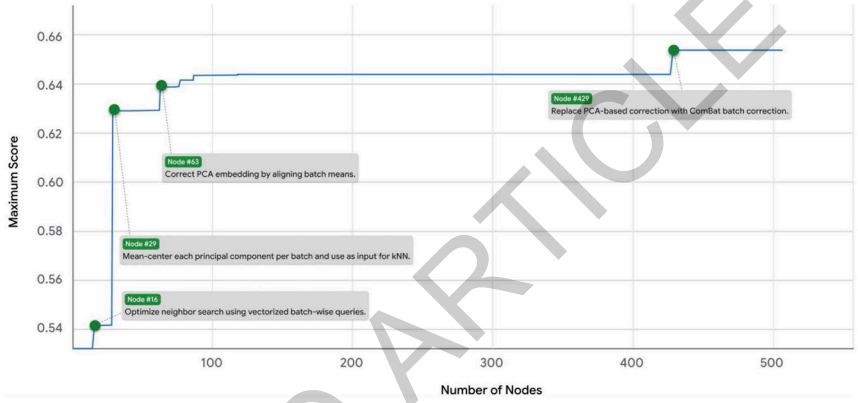
scGPT (zero shot)
ComBat x scVI
mnnCorrect (TS)
scVI x mnnCorrect
Latent OT AE
Scanorama





Extended Data Fig. 1

(A)



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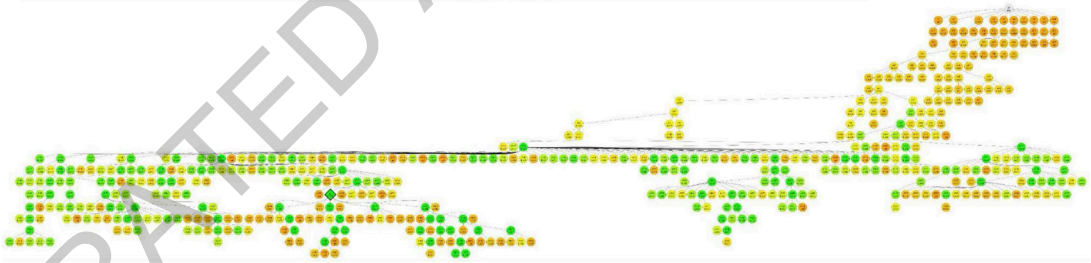

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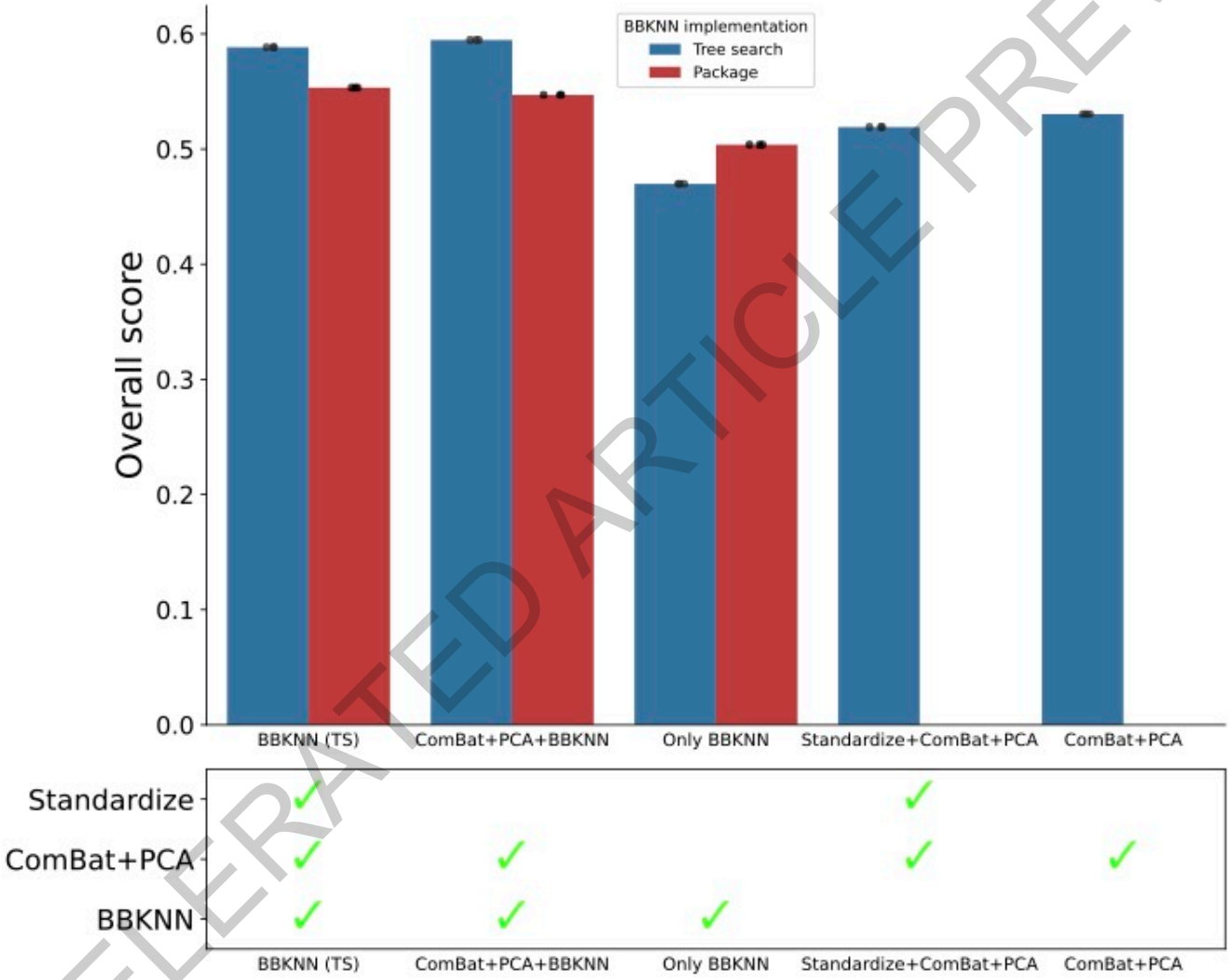
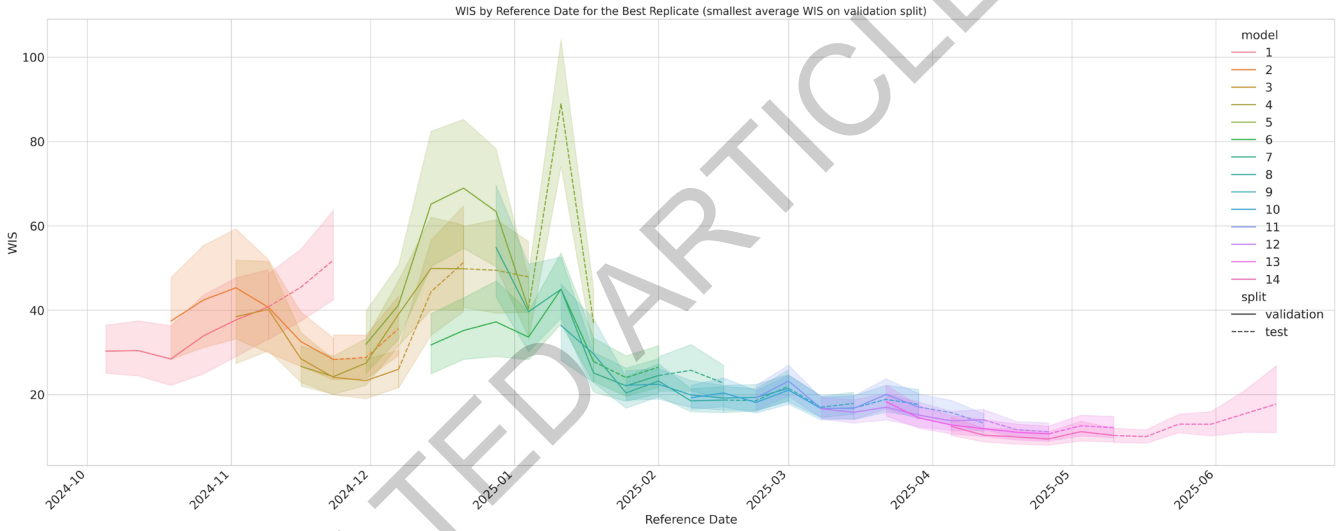


Extended Data Fig. 3

Extended Data Fig. 4



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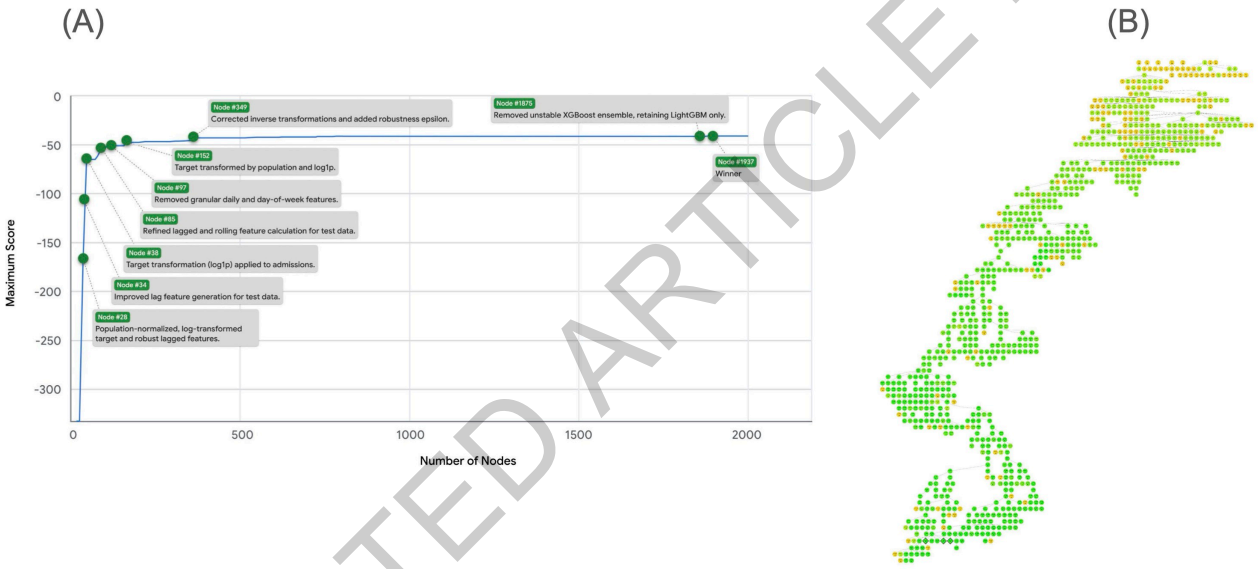

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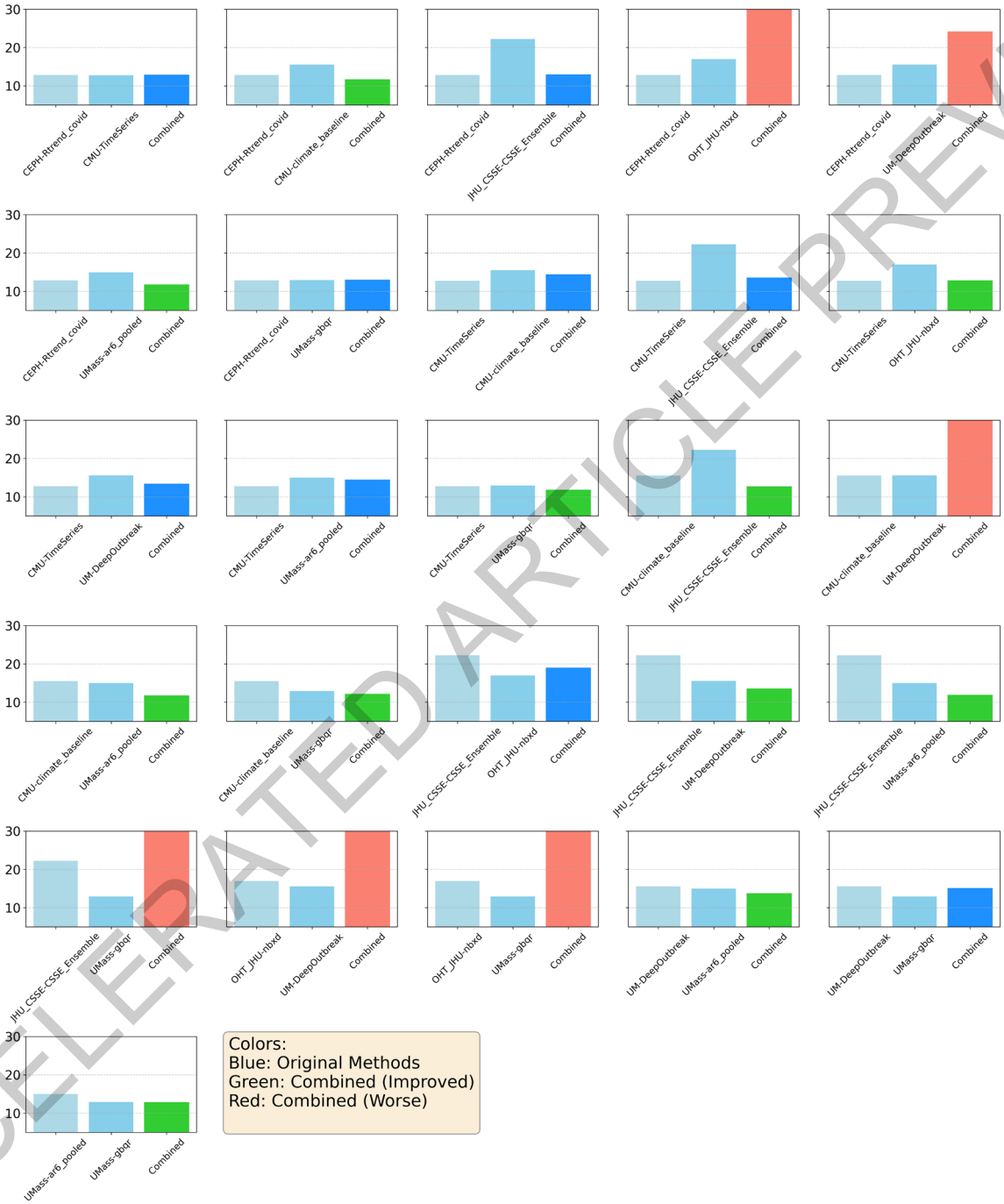
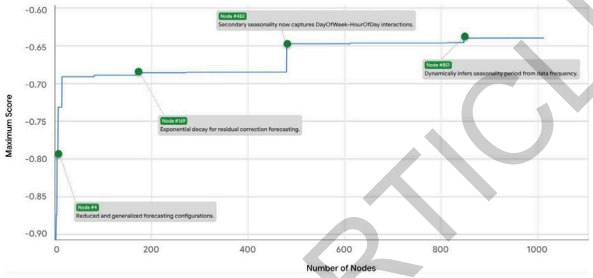


Extended Data Fig. 7

(A)



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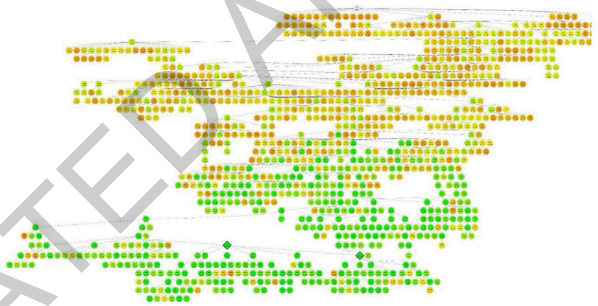

Extended Data Fig. 8